Method of expressing concentration of solution

121. (b) $\frac{X}{X+\frac{1000}{78}} = 0.2$

122. (c)$C = \frac{6}{60} = 0.1\ molar.$

123. (b)Molar solution of sulphuric acid is equal to $2N$ because it is show dibasic nature.

124. (a)$N = \frac{w\times1000}{eq.wt.\times\text{volumeinml.}}\ eq.wt. = \frac{106}{2} = 53$

$w = \frac{0.5\times53\times500}{1000} = 13.25.$

125. (c)Molar concentration = $\frac{5.85\times1000}{58.5\times200} = 0.5Molar.$

126. (c) $M = \frac{w\times1000}{m.wt.\times V\text{in}ml}$

$= \frac{75.5\times1000}{56\times540} = 2.50M$

129. (a)$N_1V_1 = N_2V_2$

$10 \times 10 = 0.1 \times$ Volume of new solution

Volume of water = 1000 – 10 = 990 $ml.$

130. (b)$W = \frac{M\times m.wt.\times V}{1000}$

$= \frac{0.1\times98\times400}{1000} = 3.92g.$

131. (a) Molarity of pure water $= \frac{1000}{18} = 55.6M$.

132. (c) $M = \frac{N}{2} = \frac{0.2}{2} = 0.1M$

133. (c) Moles of water $= \frac{180}{18} = 10mole$.

134. (c) Mole fraction of $CO_2 = \frac{n_{CO_2}}{n_{CO_2}+n_{N_2}}$

$$= \frac{\frac{44}{44}}{\frac{44}{44}+\frac{14}{28}} = \frac{2}{3}.$$

136. (c) $M = \frac{w}{m \times V(l)} \Rightarrow 0.1 = \frac{w \times 4}{40 \times 1} \Rightarrow w = 1gm$

137. (c) $M = \frac{w \times 1litre}{m.wt. \times \text{Volume } litre} = \frac{4 \times 1}{40 \times 0.1} = 1M$.

138. (b) Number of moles $= \frac{w_1}{m_1} + \frac{w_2}{m_2} = \frac{90}{18} + \frac{300}{60} = 10$

139. (a) The number of moles of solute dissolved in 1000 gm of the solvent is called molal solution.

140. (b) $w = \frac{0.1 \times 100 \times 392}{1000}$

$= 3.92g$

141. (d) $\frac{18}{180 \times 1} = \frac{1}{10} = 0.1$molal.

142. (c) $M = \frac{n}{V(l)} \Rightarrow 3 = \frac{n}{1} \Rightarrow n = 3$ moles.

143. (b)The unit of molality is mole per kilogram.

144. (a)0.2 water + 0.8 ethanol; X_A = mole fraction of water, X_B =mole fraction of ethanol

$$X_A = \frac{N_1}{N_1+N_2}, X_B = \frac{N_2}{N_2+N_1}$$

∴ Mole fraction of water = 0.2 and ethanol = 0.8.

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